Construction Permits in the Monteverde Cantón

Rebecca Stamm

Research Questions

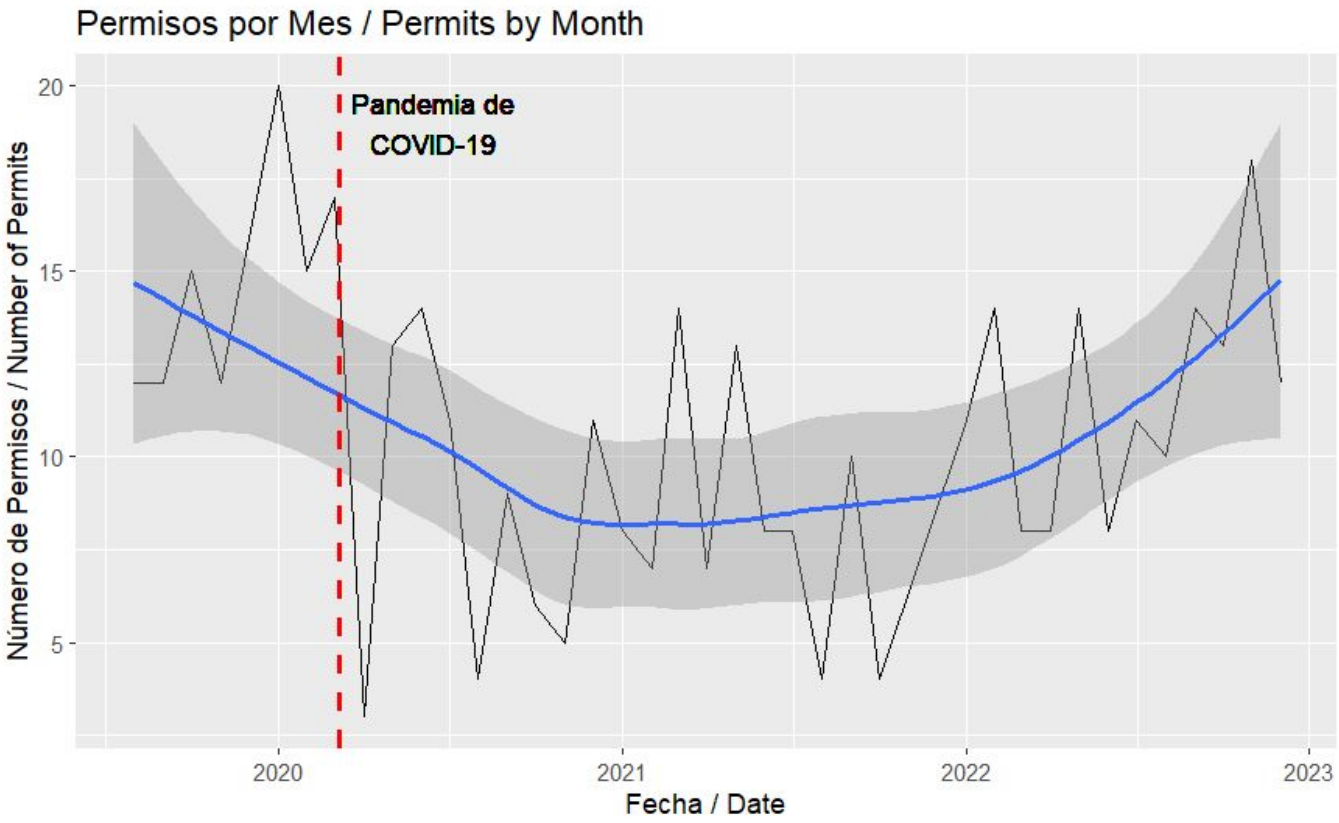
- How does the rate of construction vary depending on the season?

- What areas of Monteverde are seeing the most construction depending on the year, and what types of construction is occurring the most?

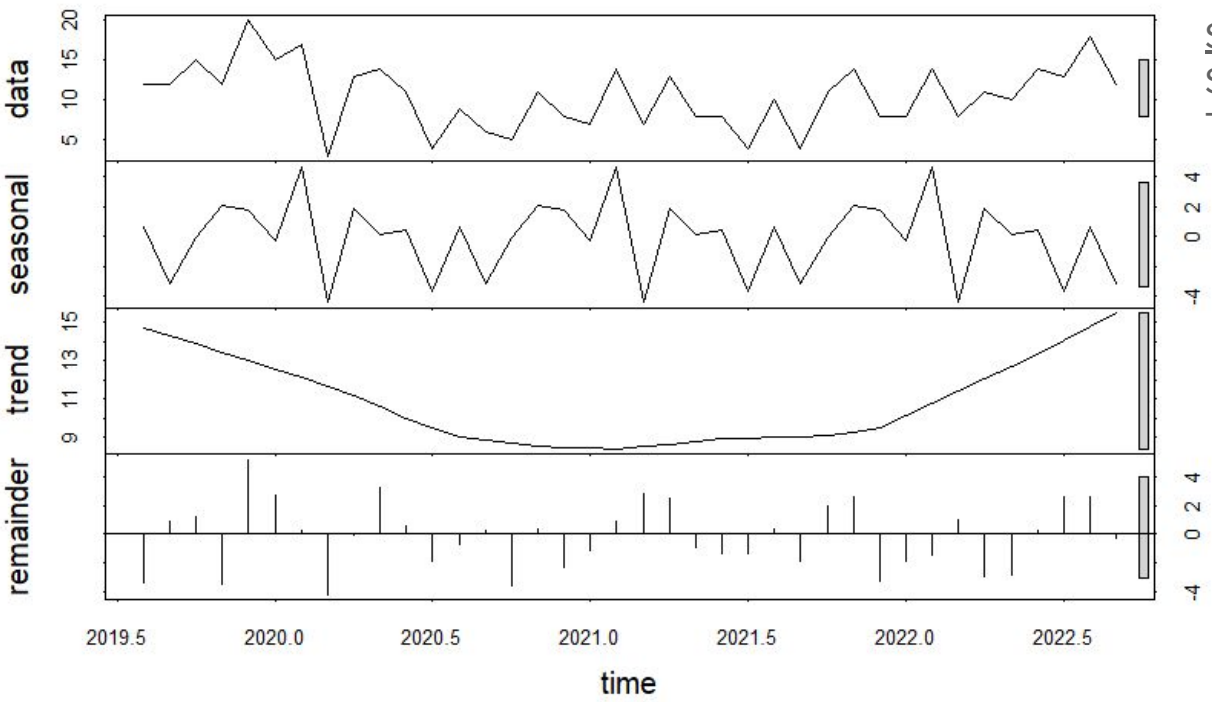
The Data

- 403 construction permits from 2019-2022
- Important variables:
  - Fecha Pago (Payment date)
  - Destino (Use / Permit Type)
  - Dirección (Address)
  - Coordenadas (Coordinates)
  - Tipo Obra (Type of Work)

The number of construction permits declined throughout the COVID-19 Pandemic, but has since recovered, as of 2022.



There is a moderately strong seasonal component with higher permit frequencies from November to February (dry season) and lower frequencies in other months.

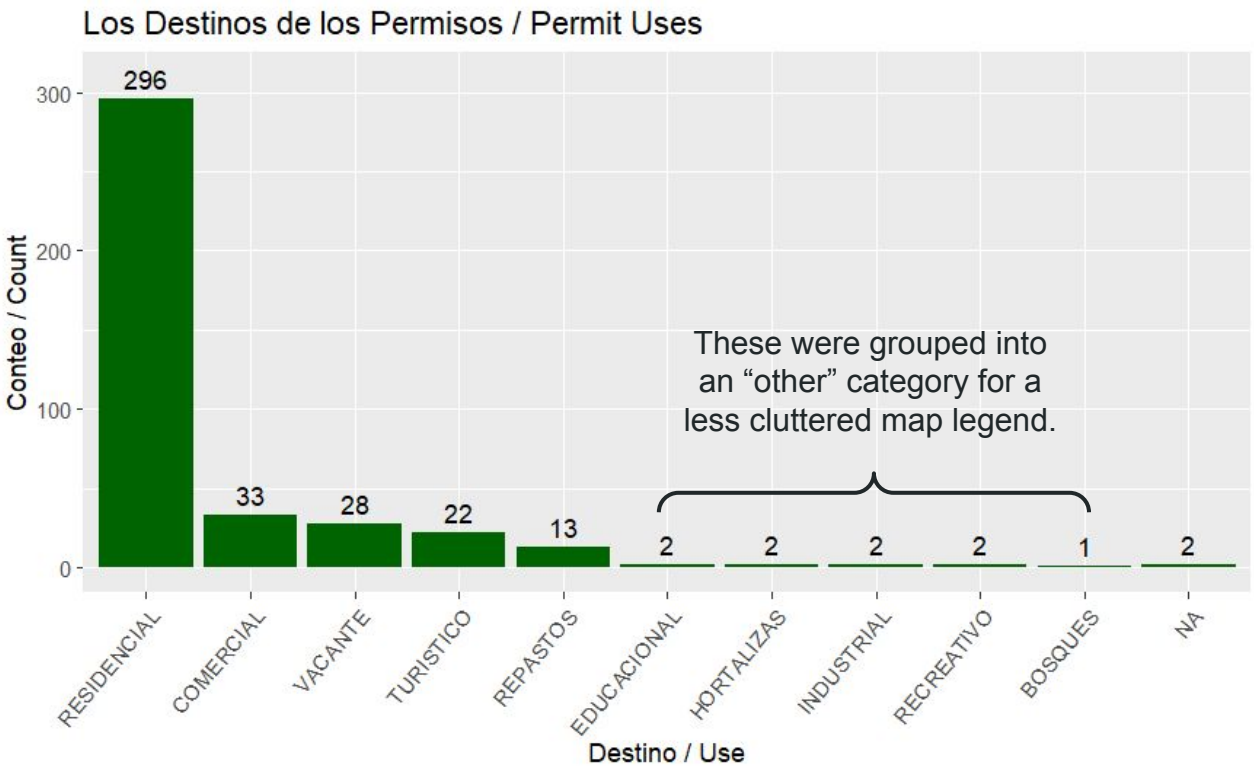


<u>Strengths</u>

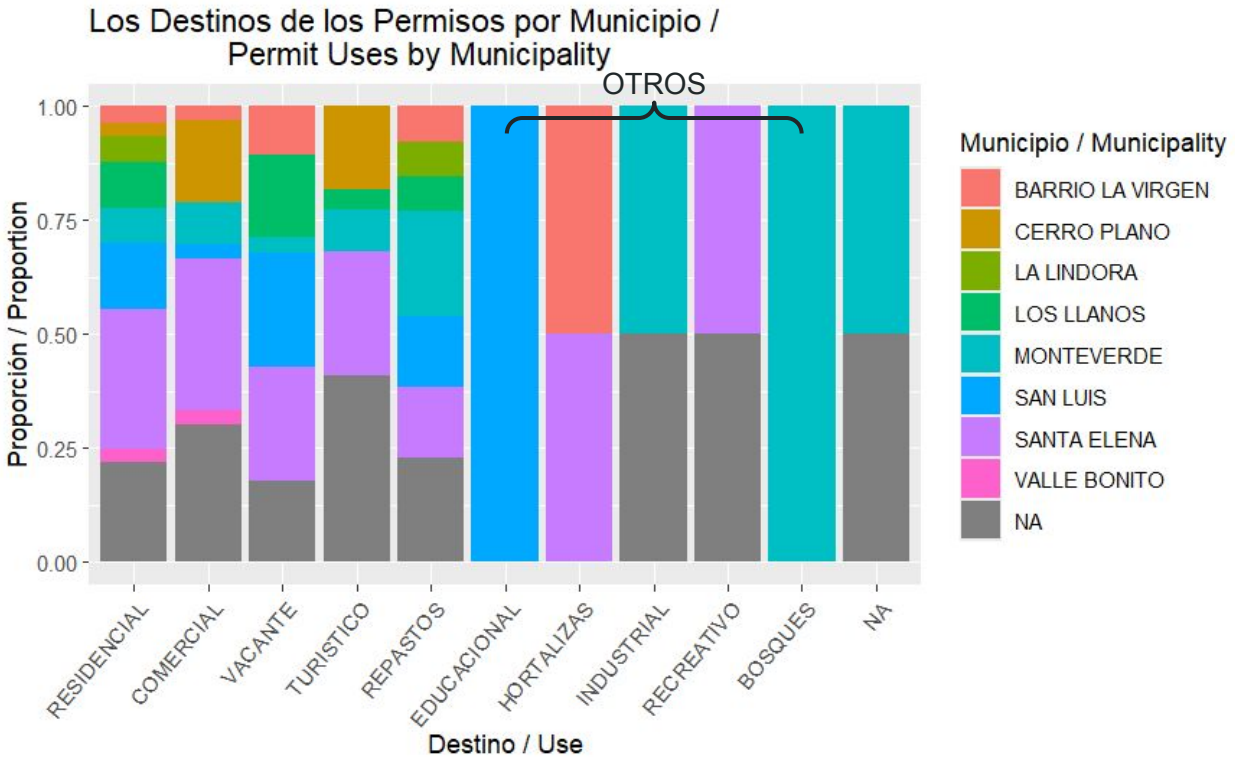
Seasonality: 0.5442

Trend: 0.4434

The majority of permits are residential, with some commercial, vacant, tourist, and pasture/farm lots.

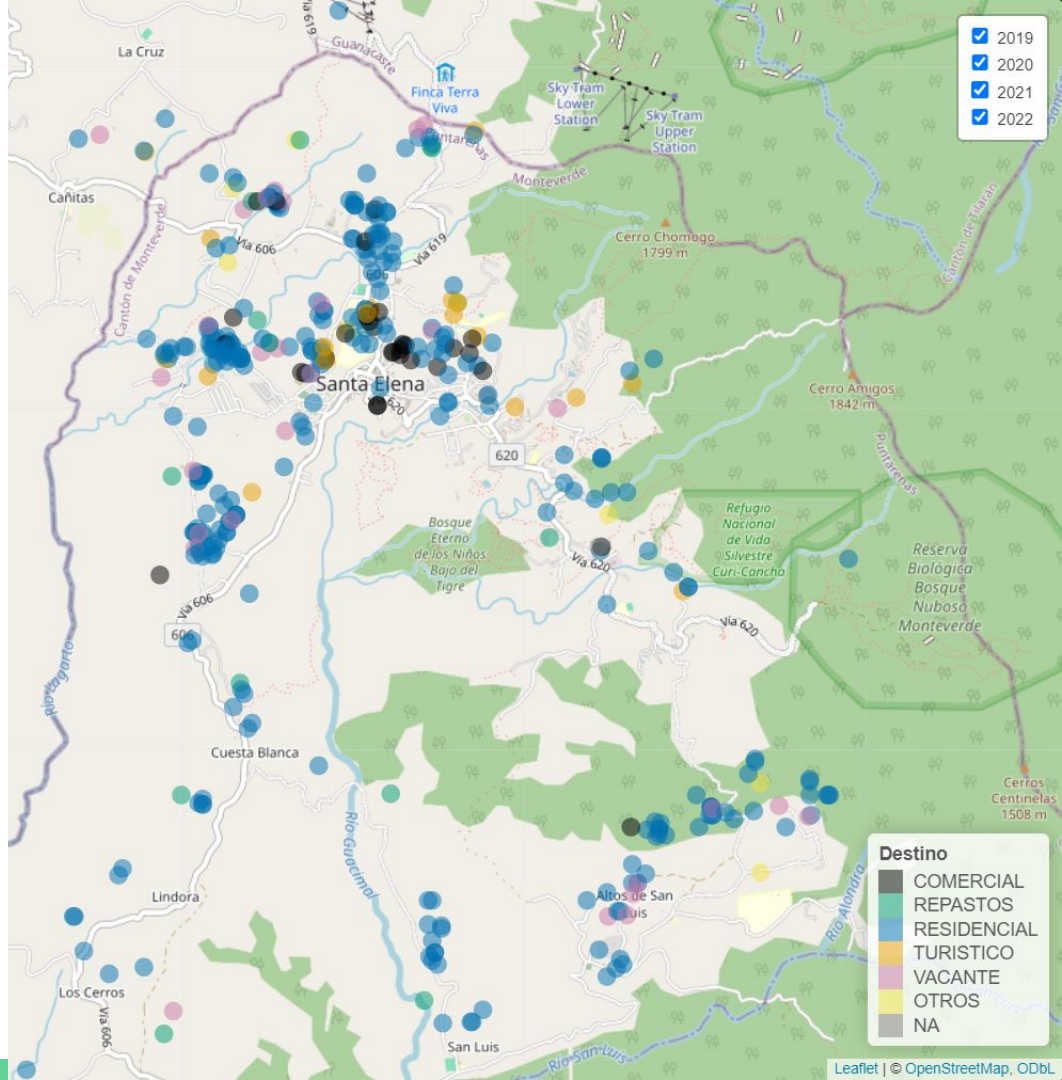


Residential permits are everywhere; commercial are mostly in Santa Elena and Cerro Plano; vacant are in Santa Elena, San Luis, and Los Llanos; tourist are in Cerro Plano and Santa Elena.

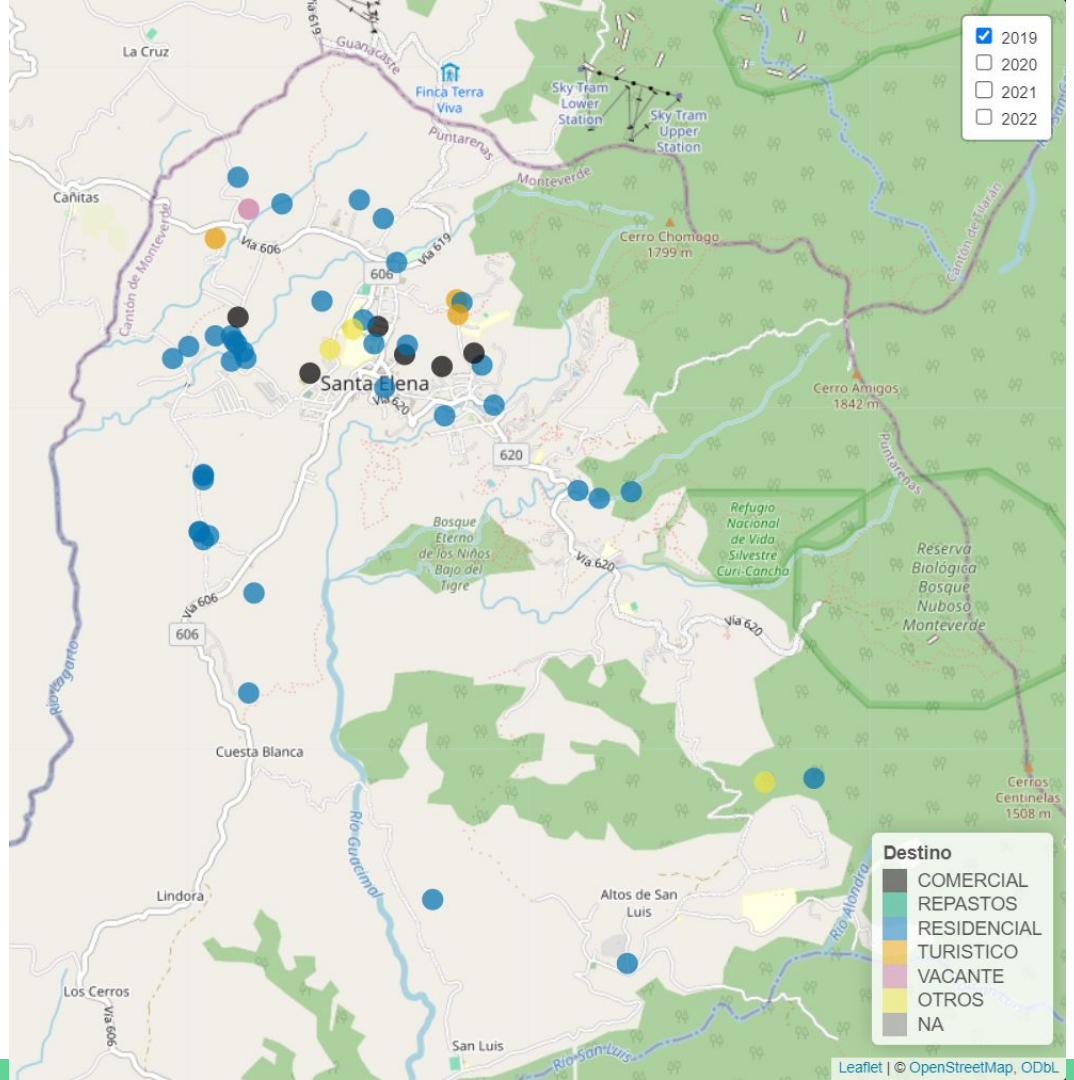


The distribution of construction permits based on use:

- Commercial and tourism properties mostly located in Santa Elena
- Residential clusters in Las Huertas, Valle Bonito, Los Llanos, and San Luis
- Mix of permit types in Barrio La Virgen
- Not as much dense construction in Cerro Plano, Monteverde Proper and La Lindora



2019


2019
2020
2021
2022
Destino
COMERCIAL
REPASTOS
RESIDENCIAL
TURISTICO
VACANTE
OTROS
NA
Santa Elena
La Cruz
Cañitas
Finca Terra Viva
Sky Tram Lower Station
Sky Tram Upper Station
Puntarenas
Monteverde
Cerro Chomogo 1799 m
Cerro Amigos 1842 m
Cantón de Monteverde
Cantón de Flacón
Via 606
Via 619
Via 620
606
620
Bosque Eterno de los Niños Bajo del Tigre
Refugio Nacional de Vida Silvestre Curi-Cancha
Reserva Biológica Bosque Nuboso Monteverde
Cerros Centinelas 1508 m
Cuesta Blanca
Lindora
Los Cerros
Altos de San Luis
San Luis
Río Guacimal
Río Alondra
Río San Luis
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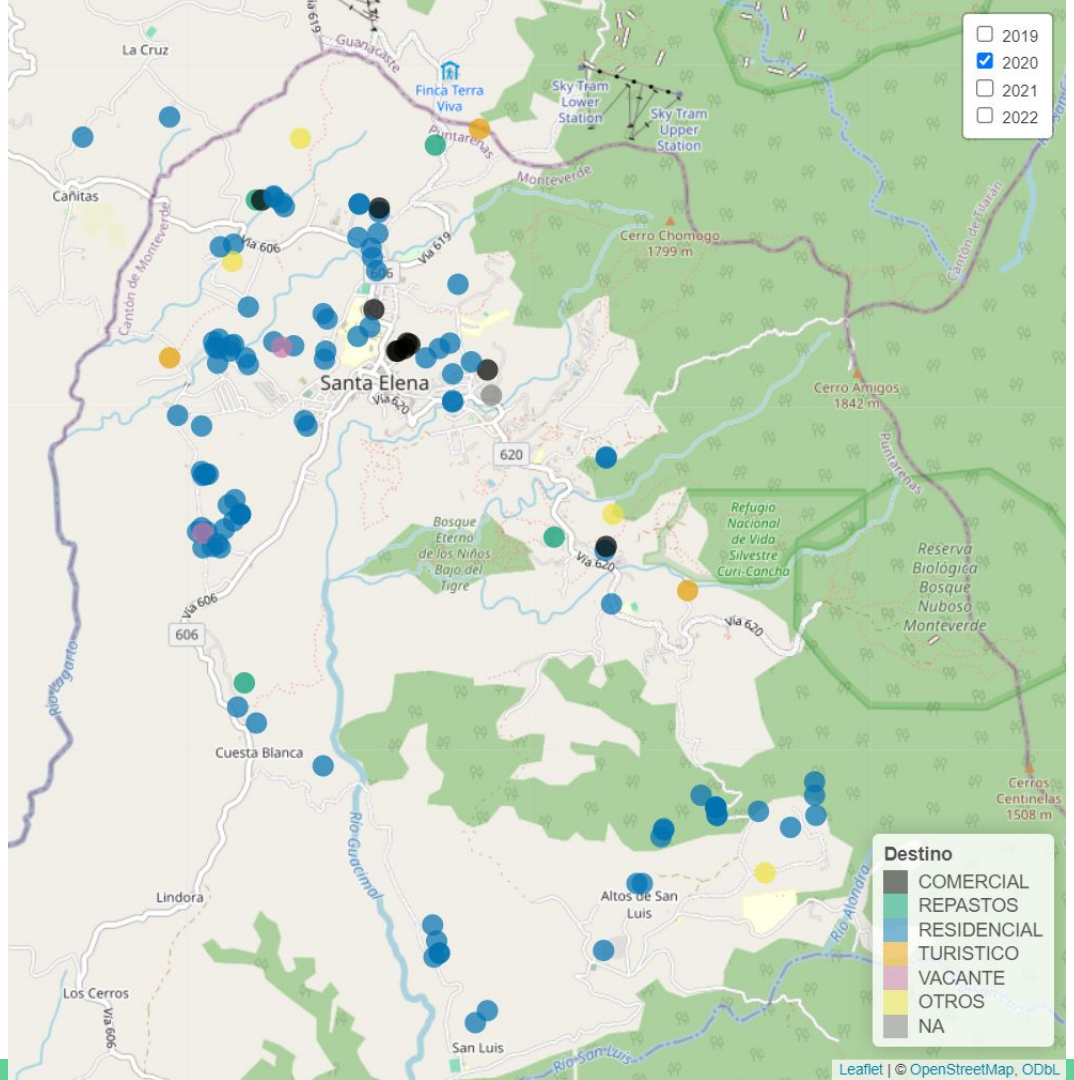
2020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Cantón de Monteverde
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OTROS
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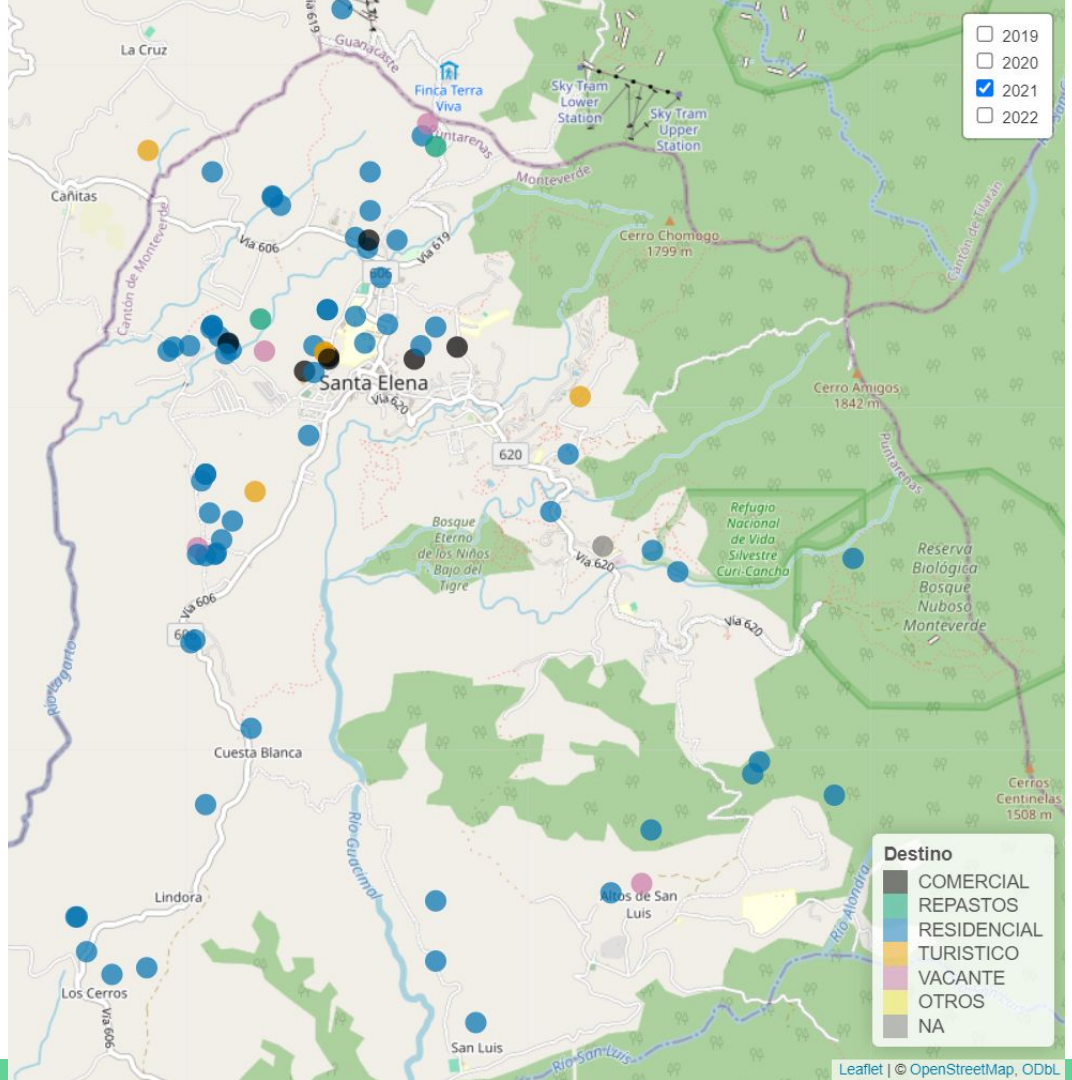
2021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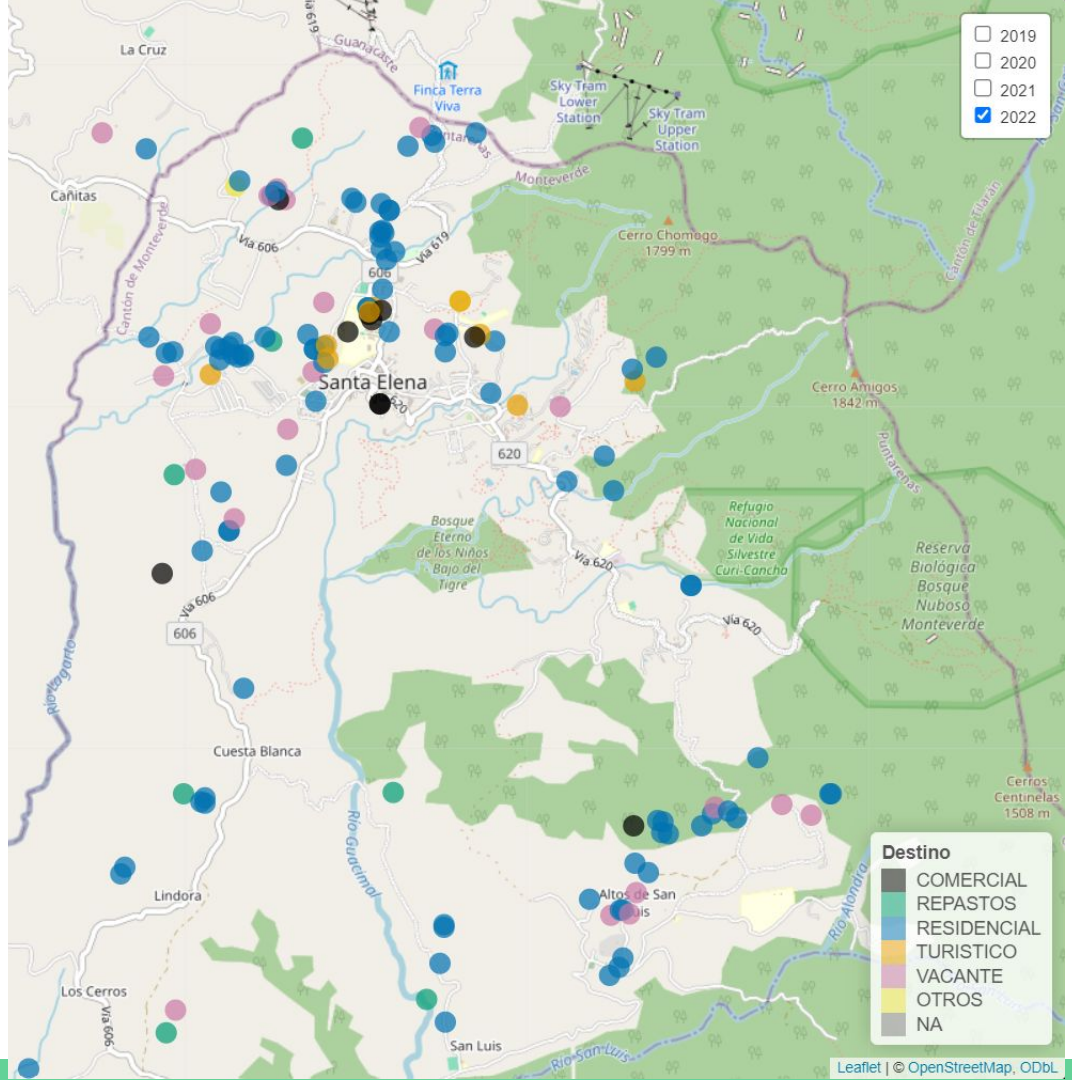
2022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Conclusions

- **Rate of Permits**
  - The rate of constructions permits declined beginning in mid-2020 due to the COVID-19 pandemic. As of 2022, the rate of permits was recovering.
  - There is a seasonal component to the rate of permit creation, with more construction permits in the dry season than in the rainy season.

- **Spatial Distribution of Permits**
  - Most of the commercial construction is in Santa Elena.
  - There are dense clusters of residential construction all over the cantón.
  - Monteverde proper and La Lindora are not seeing much dense construction.

Limitations

- Need more recent data (2023-2025)
  - To better analyze a seasonal trend without the external factor of the COVID-19 pandemic
- Incomplete addresses and coordinates
  - Many of the locations in the dataset are incomplete or missing which minimizes the ability to properly analyze relationships between the municipality or relative location and other factors.

Future Work

- Additional variables in the dataset that I did not investigate
  - Building materials (walls, floors, roofs)
  - Number of rooms and floors
  - Square meterage
  - Cost/value
  - Property owner
  - Specific type of construction work

- A most succinct and clear categorical system for destino and tipo de obra
  - Allow for deeper investigation into these variables
  - Potential for spatial mapping with tipo de obra instead of destino

Questions?